Fiscal Year:	FY 2005	**Task Last Updated:**	FY 01/04/2005
PI Name:	Ferl, Robert Ph.D.		
Project Title:	Functional genomics of plant response and adaptation to low atmospheric pressure		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Developmental biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Plant Biology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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No. of PhD Candidates:	2	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA ARC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
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COI Name (Institution):	Paul, Anna-Lisa (University of Florida)		
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	This proposal seeks a fundamental understanding of the mechanisms involved in plant responses and adaptations to low pressure environments, with the ultimate goal of producing plants that are specifically designed to thrive in low pressures. The essential drivers of this project are 1) the inescapable engineering limitations to producing orbital, lunar or Martian plant growth facilities that contain earth-normal atmospheric pressures, and 2) the knowledge that plants in certain circumstances can live under reduced pressures. In the initial phases of the project, we propose to use the tools of functional genomics in the model plant system of arabidopsis to examine molecular genetic responses that occur during initial and chronic exposure to atmospheres ranging from near earth normal to near Martian pressures and containing various gas mixtures. Based on the results from these analyses, we will identify genes and metabolic pathways that are activated and repressed in response to low-pressure atmospheres, and predict alterations that might enhance the adaptation process. Finally, we will select and directly engineer genetic changes that will test these predictions and		

Task Description:	potentially lead to plants with enhanced abilities to live in low-pressure environments. The long-term goal of this project is an understanding of the molecular changes in gene activity that accompany the response and adaptation of plants to low atmospheric pressures. Such an understanding will lead to hardware engineering and/or plant genetic engineering modifications that will mitigate the negative impacts of low-pressure atmospheres on plant growth and development, enhance adaptation and strengthen the role of plants as integral and efficient components of extraterrestrial human life support systems. There are four specific aims: 1. determine the spectrum of genes activated and repressed by low pressure 2. develop an understanding of the metabolic pathways involved in response and adaptation 3. determine the effects of defined gas mixtures in mitigating the impacts of low pressure atmospheres 4. develop and test biological engineering strategies to enhance growth in low atmospheric pressures The long-term goal of this project is an understanding of the molecular changes in gene activity that accompany the response and adaptation of plants to low atmospheric pressures. Such an understanding will lead to hardware engineering and/or plant genetic engineering modifications that will mitigate the negative impacts of low-pressure atmospheres on plant growth and development, enhance adaptation and strengthen the role of plants as integral and efficient components of extraterrestrial human life support systems.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	An understanding of plant responses to extreme environments at the molecular genetic level contibutes to basic molecular biology research.
Task Progress:	Preliminary experiments set the stage for the design of the current proposal. Experiments utilizing plants of various ages exposed to low pressure and low oxygen environments have begun to fulfill specific aim 1 and address aims 2 and 3. An experiment utilizing 9 day old plants grown vertically on nutrient agar media was designed to evaluate the effects of hypobaric (10 kPa) and hypoxic (2% oxygen) environments on global gene expression patterns. These experiments demonstrate that Arabidopsis plants respond to hypobaria with extensive changes in gene expression patterns, indicating that hypobaria elicits a dynamic adaptive response. Some of the altered gene expression patterns are similar to those involved in the hypoxic stress response, as would be expected from the low oxygen partial pressures present in hypobaria; yet some changes in gene expression patterns are unique to hypobaric conditions. These unique differential gene expression patterns demonstrate that hypobaria is not equivalent to hypoxia as an abiotic stress and suggest that, for plants, the response to hypobaria is more complex than the acclimation to the reduced partial pressures of oxygen inherent to low atmospheric pressures. By extension, environmental compensation for hypobaric stress in plants is not as simple as increasing the oxygen content as was done for humans in the low pressure environments of the early space age. These data were published in the journal Plant Physiology (cover article) in January 2004 (Paul et al., 2004). The design of the next series of experiments builds on these findings. They are described briefly below, along with their respective status of analysis. 1. Variations in hypobaric stress – mature arabidopsis plants. These experiments were designed to evaluate the effects of exposing plants to hypobaric environments of increasing severity. Plants were exposed to atmospheres of 101 kPa (control – normal atmosphere at sea level), 75 kPa, 50 kPa, 25 kPa and 10 kPa for 24 hours then harvested for RNA extraction and analyses. RNA has been prepared for all of these samples and they have been evaluated for global changes in gene expression patterns with Affymetrix microarray gene chip technologies (see Paul et al. 2004 for details). The data from the microarrays have been analyzed and these data are in the process of being developed for publication. 2. Variable atmospheres and developmental stages – plate grown arabidopsis plants. These experiments were conducted at the University of Guelph in collaboration with Dr. Mike Dixon. This set of experiments was extensive, and designed to be comprehensive enough to address the first three aims, and thereby lead to the development of strategies for aim 4. The experiments were conducted throughout the month of May in Guelph and are comprised of five parts. 2 - Experiment 1 – week of May 3rd: variation in pressure, 24 hr exposure, 11 day plate plants. Total plates needed for Exp 1 = 225. Plants for this experiment will be planted on 4/15 then placed at 4° C. Growth will be initiated when the plates are removed from the cold on designated days by Mike's group: Exp 1: 101, 75, 10 set of plates placed in 23° C growth chamber on 4/23; Exp 1: 25, 50, 5 set of plates placed in 23° C growth chamber 4/24 and Exp 1: Ox1, Ox2*, 0.7 set of plates placed in 23° C growth chamber 4/25. Exp 1, Day 1 (Tue 5/4) - 11 day old plate-grown plants will be exposed to 101, 75, 10 in triplicate, plus TC callus tissue in 101 and 50 treatments. Each treatment set is loaded with a 60 minute stagger to help facilitate the harvest step at the end of the 24 hour period. Three internal replicates composed of 3 plates separated into roots and shoots (R, S) will be collected from each chamber. Exp 1, Day 2 (Wed 5/5) - 11 day old plate-grown plants will be exposed to 25, 50, 5 in triplicate, plus TC callus tissue in 10 and 5 treatments. Exp 1, Day 3 (Thur 5/6) - 11 day old plate-grown plants will be exposed to Ox1, Ox2, 0.7 in triplicate Status: Most of the samples from this experiment are being stored for future use (in RNAlater at -20°C). From this set, only the isuue culture samples have been fully processed. The tissue culture RNAs were submitted for microarray analyses with the samples from Experiment 2.2 (below). 2 - Experiment 2 – week of May 10th: variation in atmospheric pressure, 24 hr exposure, 10 day and 5 day old plants for arrays. Total plates needed for Exp 2 = 720. 10 day old plate-grown plants will be represented with sufficient abundance for array analyses (internal replicates composed of three plates) while in the younger plant (5 day old) replicates are composed of one plate each (providing 3 internal replicates). All plates/ages will be exposed to 9 different atmospheric pressures for a period of 24 hours: 101 kPa, 75 kPa, 50 kPa, 25 kPa, 10 kPa, 5 kPa, Ox1, Ox2, Ox3. Nine chambers are used per day in treatment sets of three replicates. Each treatment set is loaded with a 60 minute stagger to help facilitate the harvest step at the end of the 24 hour period. Three biological replicates composed of 3 plates each (9 total) separated into roots and shoots (R, S) will be collected from each chamber for the 10 day old plants. The younger plants will be represented by four plates (1 plate for each biological replicate) but the plates will be more thickly planted, and in several tiers (dependent on age). 10 day plants are separated into roots and shoots and harvested to RNAlater in 2.0 mL screw-top microfuge tubes. 5 day plants are harvested whole. Exp 2, Day 1 (Mon 5/10) – 10day and 5day exposed to 101, 75, 10 in triplicate Exp 2, Day 2 (Tue 5/11) – 10day and 5day exposed to 25, 50, 5 in triplicate Exp 2, Day 3 (Wed 5/12) – 10day and 5day exposed to Ox1, Ox2, Ox3 in triplicate Ox1 = 50kPa with O2 added to equal normoxic conditions Ox2 = 25kPa with O2 added to equal normoxic conditions Ox3 = 101kPa but with a reduced oxygen atmosphere to equal pp at 25kPa (5% Oxygen) Status: The RNA has been isolated from the samples described in this experiment and have been submitted for microarray analyses. The analysis of the array data is on-going. 2 - Experiment 3 – week-end of May 13th: Development under pressure. Total plates needed for Exp 3 = 200. A series

	of ages (0, 3, 5, 8, 10 day old) will be started in the chambers together and allowed to grow in the designated atmosphere (101, 50, 10) for 72 hours. All harvested whole to RNAlater beginning Monday 5/17 Status: The samples from this experiment are being stored for future use (in RNAlater at -20°C). 2 - Experiment 4 – wk of May 17th: Exposure time course. Total plates needed for Exp 4 = 180. 10 day old plate-grown plants will be exposed to 10 kPa and samples will be taken over a 72 hour time course: 72 hrs, 48 hrs, 12 hrs, 6 hrs, 3 hrs, 1 hr, plus 101 controls harvested with each time point. Six different chambers will be used on each of two separate sets of launches (Set 1 and Set 2). Three replicates composed of 3 plates each (total of 9) separated into roots and shoots (R, S) will be collected for each time point, plus corresponding controls at each time. Plants are and harvested to RNAlater. Status: The samples from this experiment are being stored for future use (in RNAlater at -20°C). 2 - Experiment 5 – 0.7 kPa (Mars pressure) time course. 10 day old plate-grown plants will be exposed to 0.7 kPa, observed and samples will be taken over a 12 hour time course: 12 hrs, 6 hrs, 3 hrs, 1 hr, plus 101 controls harvested with each time point. Two sets of chambers (6) chambers will be used. At least three replicates composed of 1 plate each (could run 4 total if you had them) separated into roots and shoots (R, S) will be collected for each time point, plus corresponding controls at each time (below). Plates should be set up with double thick agar. This can be done with an existing plate (of the appropriate age) by carefully lifting the entire agar/seedling mass off of one plate and laying it on blanket-wise to another (blank) plate of agar. Lay on smoothly with a rolling motion to avoid trapping bubbles. Status: The samples from this experiment are being stored for future use (in RNAlater at -20°C).
Bibliography Type:	Description: (Last Updated: 11/27/2022)
Articles in Peer-reviewed Journals	Paul A-L, Schuerger AC, Popp MP, Richards JT, Manak MS, Ferl RJ. "Hypobaric biology: Arabidopsis gene expression at low atmospheric pressure." Plant Physiol. 2004 Jan;134(1):215-23. PMID: 14701916 , Jan-2004
Journal/Magazine covers	Paul A-L, Schuerger AC, Popp MP, Richards JT, Manak MS, Ferl RJ. "Cover in journal Plant Physiology for article, “Hypobaric biology: Arabidopsis gene expression at low atmospheric pressure." Plant Physiol. 2004 Jan;134(1):215-23. PMID: 14701916 , Jan-2004
Presentation	A-L. Paul and R.J. Ferl "Genomic Responses of Arabidopsis to Abiotic Stresses Relevant to Mars Exploration and Colonization" Habitation 2004 meeting in Orlando Florida Jan-2004
Presentation	A-L. Paul and R.J. Ferl "Changes in Plant Gene Expression in response to reduced atmospheric pressure " Mars on Earth Conference, Santa Fe NM Mar-2004
Presentation	R.J. Ferl "Life to Mars - A Biologist's perspective on extraterrestrial habitation" Ethel Belk Lecture in Botany at the Miami University - Mar-2004
Presentation	A-L. Paul: "Hypobaric biology: Arabidopsis gene expression at low atmospheric pressure" University of Florida, Horticultural Sciences Department fall seminar series - Nov-2004
Presentation	R.J. Ferl "Genomic Responses of Arabidopsis to Abiotic Stresses Relevant to Mars Exploration and Colonization" NASA/SLSL seminar series Dec-2004
Significant Media Coverage	Sikes JY. "Crops for Astronauts, coverage about programs associated with R.J. Ferl, A-L. Paul and colleagues." Gainesville Sun Newspaper, December 2004., Dec-2004